

DAY 7 – PYTHON STRINGS (DETAILED NOTES)

What is a String?

A string is a sequence of characters enclosed in single quotes (' '), double quotes (" "), or triple quotes. Strings are immutable, meaning they cannot be changed after they are created. Every string is an object of the **str** class.

Syntax:

```
name = "Python"
```

strip()

Definition: Removes spaces from both the beginning and the end of a string.

Syntax:

```
string.strip()
```

Example:

```
text=' Python '  
print(text.strip())
```

Output: Python

lower()

Definition: Converts every uppercase letter into lowercase.

Syntax:

```
string.lower()
```

Example:

```
text='PyThOn'  
print(text.lower())
```

Output: python

upper()

Definition: Converts every lowercase letter into uppercase.

Syntax:

```
string.upper()
```

Example:

```
text='python'  
print(text.upper())
```

Output: PYTHON

lstrip()

Definition: Removes spaces only from the left side.

Syntax:

```
string.lstrip()
```

Example:

```
text=' Python '  
print(text.lstrip())
```

Output: Python

rstrip()

Definition: Removes spaces only from the right side.

Syntax:

`string.rstrip()`

Example:

```
text='Python '  
print(text.rstrip())
```

Output: Python

replace()

Definition: Replaces one word or character with another.

Syntax:

`string.replace(old,new)`

Example:

```
text='I like Python'  
print(text.replace('Python','Java'))
```

Output: I like Java

join()

Definition: Joins elements of an iterable into one string.

Syntax:

`separator.join(iterable)`

Example:

```
names=['Adi','Riya','Sam']  
print(' - '.join(names))
```

Output: Adi - Riya - Sam

find()

Definition: Returns the index of the first occurrence of a substring. Returns -1 if not found.

Syntax:

`string.find(substring)`

Example:

```
text='Programming'  
print(text.find('gram'))
```

Output: 3

startswith()

Definition: Checks whether a string begins with a given prefix.

Syntax:

`string.startswith(prefix)`

Example:

```
text='Python'  
print(text.startswith('Py'))
```

Output: True

endswith()

Definition: Checks whether a string ends with a given suffix.

Syntax:

```
string.endswith(suffix)
```

Example:

```
text='Python'  
print(text.endswith('on'))
```

Output: True

split()

Definition: Splits a string into a list.

Syntax:

```
string.split(separator)
```

Example:

```
text='Python Java C'  
print(text.split())
```

Output: ['Python','Java','C']

capitalize()

Definition: Converts only the first character to uppercase.

Syntax:

```
string.capitalize()
```

Example:

```
text='python programming'  
print(text.capitalize())
```

Output: Python programming

title()

Definition: Converts the first letter of every word to uppercase.

Syntax:

```
string.title()
```

Example:

```
text='python programming language'  
print(text.title())
```

Output: Python Programming Language

Important Points

- Strings are immutable.
- Indexing starts from 0; negative indexing starts from -1.
- Slicing syntax: string[start:end:step].
- Most string methods return a new string instead of modifying the original.

Practice Programs

1. Take a sentence and remove extra spaces.
2. Convert user input to uppercase and lowercase.
3. Replace all spaces with underscores.
4. Check if an email ends with '.com'.

5. Split a full name into first and last names.
6. Join a list of subjects using commas.
7. Convert a sentence to title case.

Viva Questions

1. What is a string?
2. Why are strings immutable?
3. What is the difference between strip(), lstrip(), and rstrip()?
4. What is the difference between find() and replace()?
5. What does split() return?
6. What is the use of join()?
7. What is the difference between capitalize() and title()?